OpenClaw Temperature and Humidity Dashboard Tutorial

> This article assumes that the Jetson Nano Docker environment, OpenClaw expansion board, DHT, OLED, RGB, servo, and three dialogue methods are already configured. This article only covers the temperature and humidity dashboard application.
>
> Features in this tutorial may require training. If command control fails, you can first describe the operation process in detail, then refine the dialogue.

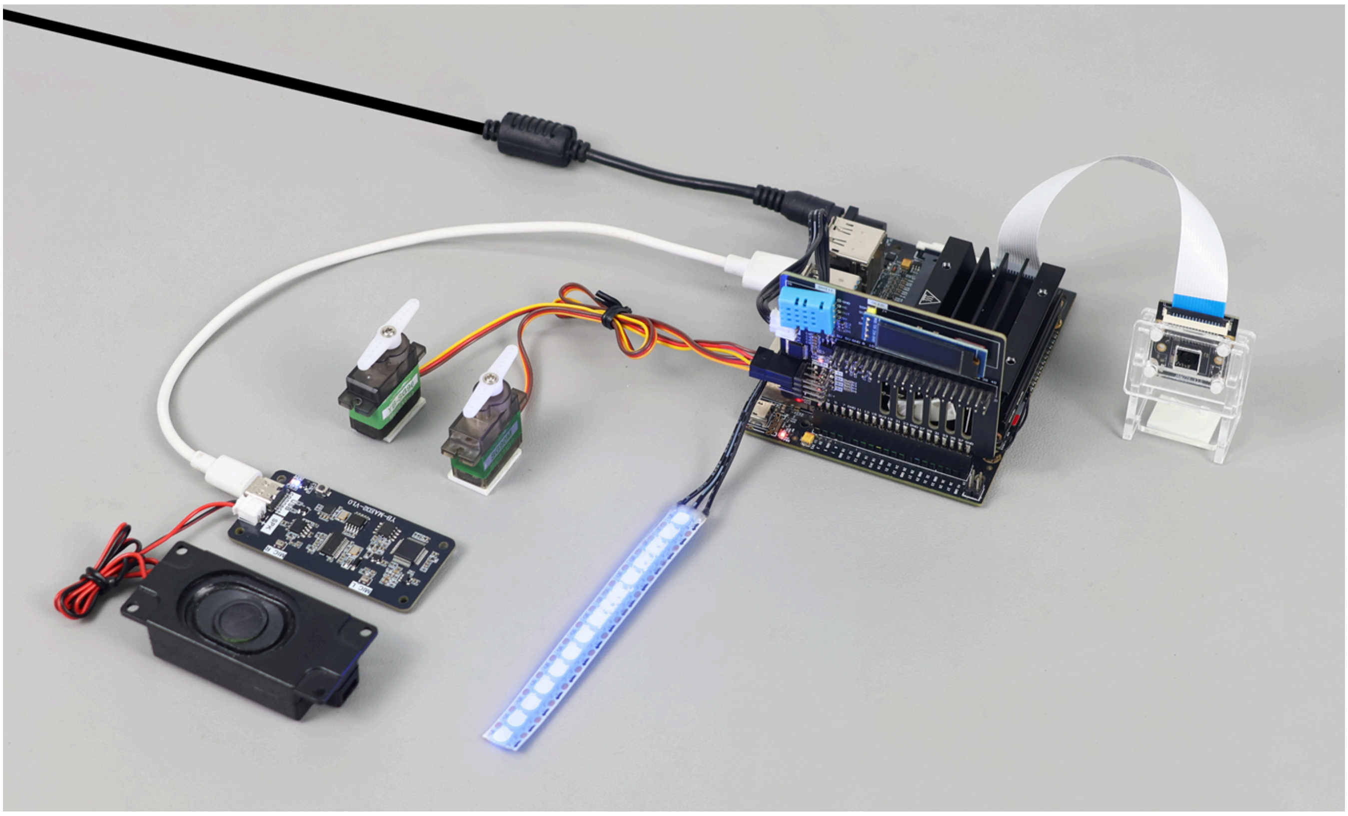
1. Tutorial Objectives

The temperature and humidity dashboard project transforms environmental data into visual feedback through the combination of "reading temperature and humidity + OLED display + RGB status light + servo pointer". After completion, you can:

- Get current temperature and humidity.
- Display temperature and humidity on OLED.
- RGB shows blue, yellow, and red lights based on temperature status.
- Two servos simultaneously rotate to the angle corresponding to temperature (angle = (tem ÷ 50) × 180 = tem × 3.6).
- Support TUI, WhatsApp, and voice triggers.

2. Hardware Preparation

- Jetson Nano B01
- OpenClaw expansion board
- RGB light strip
- Servo x 2
- Optional: AI voice interaction module + speaker
- Wiring diagram



3. Software Entry

Enter the Jetson Docker container:

```
~/run_usb_docker.sh
```

Common commands:

```
/usr/local/bin/dht --format json
/usr/local/bin/dht_oled --format json
/usr/local/bin/servo call-tool ServoAll --action set_all --angle1 90 --angle2 90
/usr/local/bin/rgb --format json color --color blue
```

Dashboard rules:

- Read current temperature, for example 29 degrees.
- Servo angle = temperature × 3.6, for example 25 × 3.6 = 90 degrees.
- Servos simultaneously rotate to that angle.
- Low temperature lights blue, comfortable temperature lights yellow, high temperature lights red.

4. Basic Capability Testing

Enter TUI:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui

🦞 OpenClaw 2026.3.8 (3caab92) — I'll do the boring stuff while you dramatically stare at the logs like it's cinema.

 openclaw tui - ws://127.0.0.1:18789 - agent main - session main

 session agent:main:main
 gateway connected | idle
 agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)

█
```

Test first:

- Read current temperature and humidity and display on OLED
- Change RGB light to green
- Control both servos to rotate to 180 degrees

5. Three Dialogue Methods

Method	Suitable Scenario	Recommended Use
TUI	Local debugging of readings and servo angles	Prioritize verifying mapping rules
WhatsApp	Remote viewing of dashboard status	Suitable for phone demonstration
Voice	Desktop voice trigger	Suitable for one-sentence to start dashboard

5.1 WhatsApp Method

You can send:

- Control servo to rotate to corresponding angle based on temperature, temperature*3.6=servo angle
- Temperature and humidity dashboard

20:36

< 2 +86 153 3885 7526 (你)
给自己发消息

warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically.
20:34

Temperature and humidity dashboard
20:35

Temperature and humidity dashboard is now active!

📊 **Current Readings:**

- **Temperature:** 29.7°C (85.5°F)
- **Humidity:** 54%

📟 **OLED Display:** Showing temp and humidity values

💡 **RGB Indicator:** Red blinking light (speed 5)

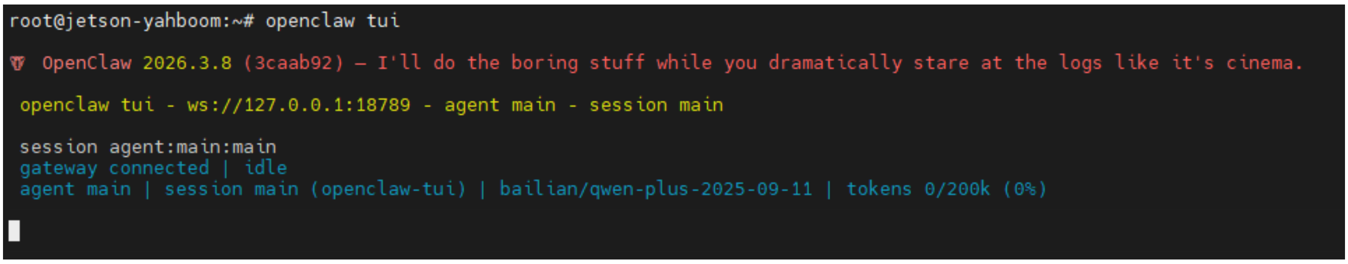
- This indicates the temperature is **above 28°C** (warm/hot zone)

Status: The room is quite warm. Consider turning on a fan or AC if you're feeling uncomfortable!
20:35

5.2 TUI Method

Run:

```
openclaw tui
```



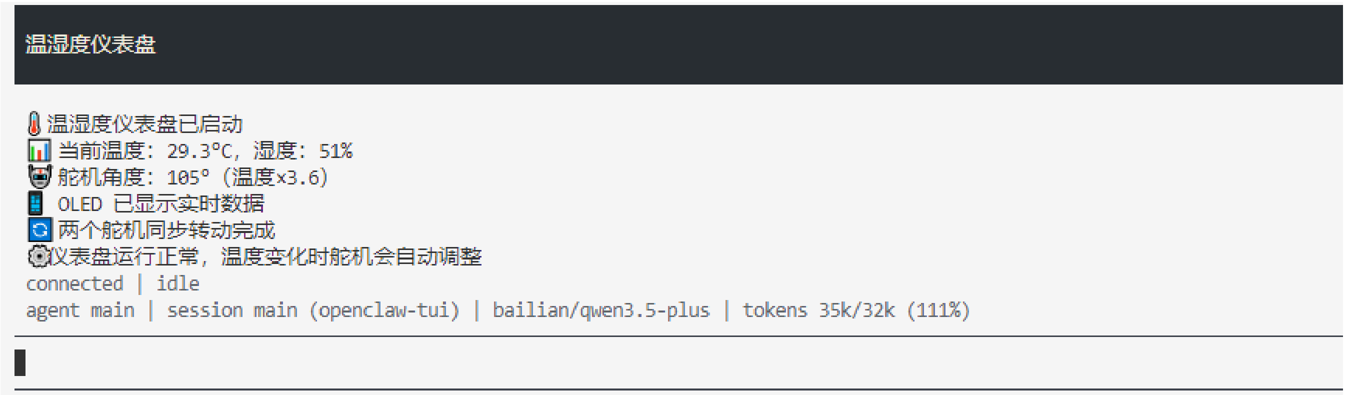
```
root@jetson-yahboom:~# openclaw tui
 OpenClaw 2026.3.8 (3caab92) — I'll do the boring stuff while you dramatically stare at the logs like it's cinema.

 openclaw tui - ws://127.0.0.1:18789 - agent main - session main

 session agent:main:main
 gateway connected | idle
 agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

Recommended inputs:

- Get temperature and humidity displayed on OLED, then control servos to simultaneously rotate to corresponding angles
- Temperature and humidity dashboard



```
温湿度仪表盘

🌡温湿度仪表盘已启动
📊当前温度: 29.3°C, 湿度: 51%
🎛舵机角度: 105° (温度×3.6)
📱OLED 已显示实时数据
🔄两个舵机同步转动完成
⚙仪表盘运行正常，温度变化时舵机会自动调整
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 35k/32k (111%)
```

5.3 Voice Method

Execute the following command while openclaw gateway is running:

```
cd /root/yahboom_ws/src/largemodel
python3 main.py
```

You can say:

- Get temperature and humidity displayed on OLED, and control servo to rotate to corresponding angle

```
root@jetson-yahboom:~/yahboom_ws/src/largemodel# python3 main.py
[main] Jetson OpenClaw voice launcher
[main] bridge ready
[main] tts ready
[main] asr ready
[main] Voice interaction is running. Press Ctrl+C to stop.

[asr] I'm here

[asr] 正在识别...... [███████████████████-] 14.8/15s
[user] 获取温湿度显示在led上，并控制舵机转到对应的角度。
[assistant] 温度 29.7 度、湿度 50% 已显示，舵机转到 107 度啦！
```

6. Comprehensive Cases

6.1 Real-time Reading

Experiment objective: Read current temperature and humidity as dashboard input.

Example commands:

- Read current temperature and humidity
- What is the current temperature and humidity

6.2 OLED Data Panel

Experiment objective: Display temperature and humidity on OLED.

Example commands:

- Display temperature and humidity on OLED
- Make a simple temperature and humidity display page

6.3 RGB Status Light

Experiment objective: Use RGB to indicate temperature status.

Example commands:

- Light red if temperature is high
- Low temperature lights blue, comfortable temperature lights yellow, high temperature lights red

6.4 Servo Pointer Mapping

Experiment objective: Map temperature to servo angle.

Example commands:

- Temperature and humidity dashboard
- After reading temperature, let both servos simultaneously rotate to corresponding angle

7. Common Problems

- Only OLED displaying, no servo action: May need to first specifically describe how to rotate the servo, then reduce statements later, or check if `servo-all` was generated.

- DHT reading failed: Execute `dht` or `dht_oled` separately to check if normal. If reading fails, it may be due to internal DHT instability, retry once.
- When running main.py, if you see `[user] Failed to open KWS serial port.` error, unplug and replug the voice module